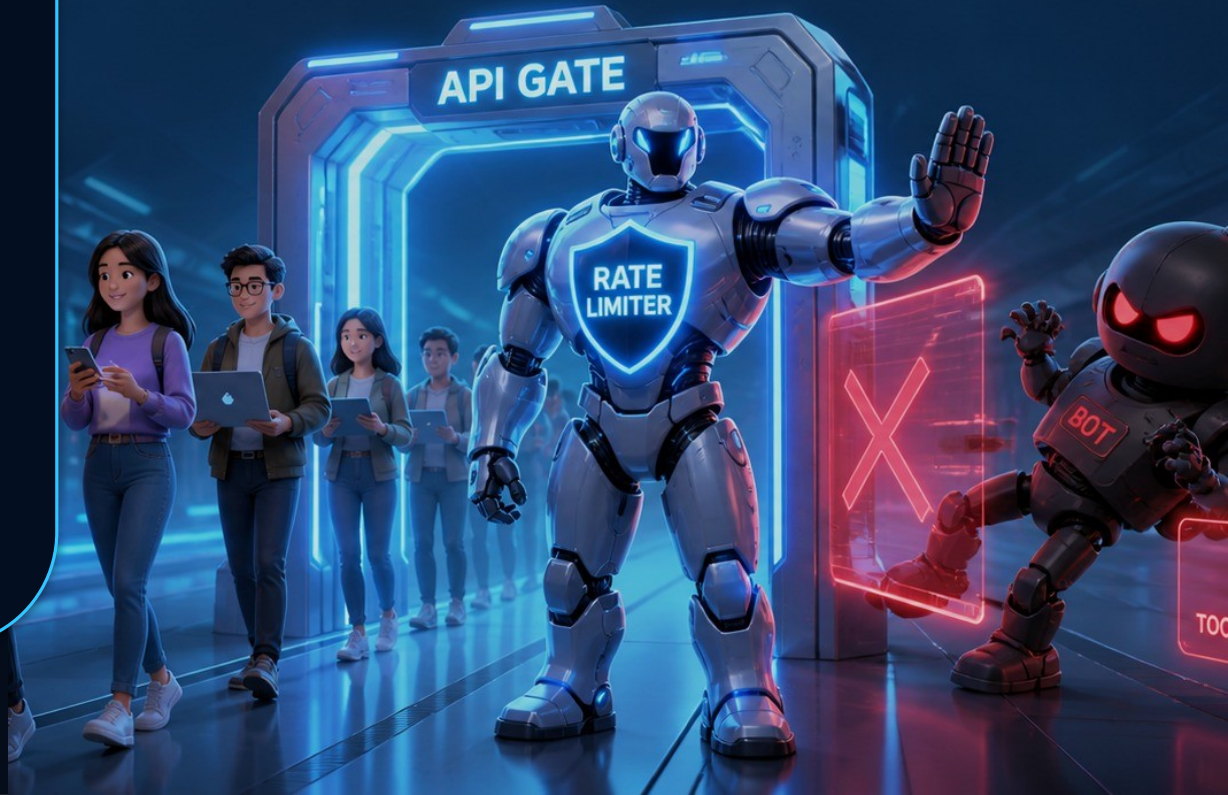


MEET LIMITER.

the bouncer
guarding your API

RATE LIMITER, DRAWN · EP 1 / 5

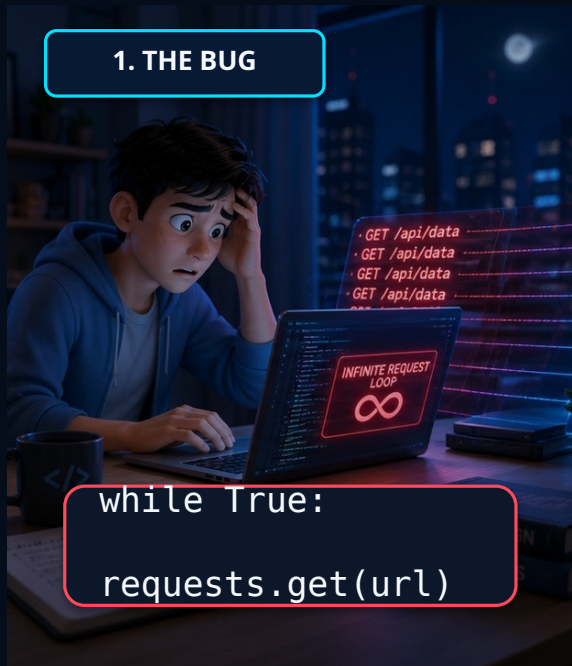
Story rule: humans pass, bots wait.



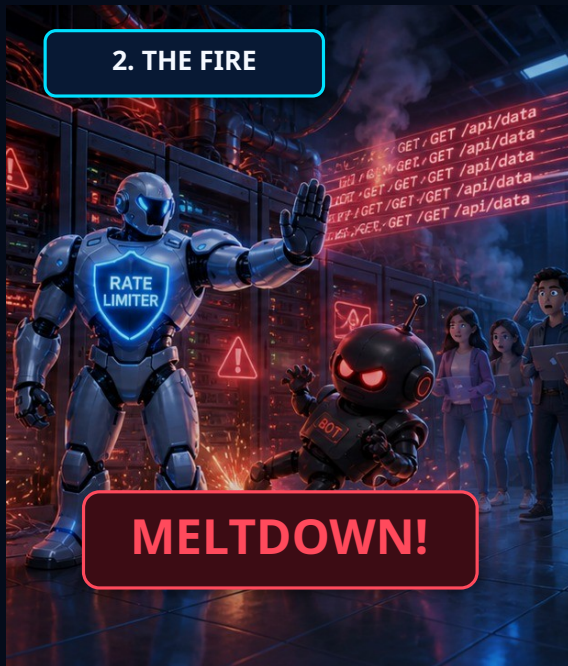
ONE BUG. ONE NIGHT. \$40K GONE.

SCENE 01

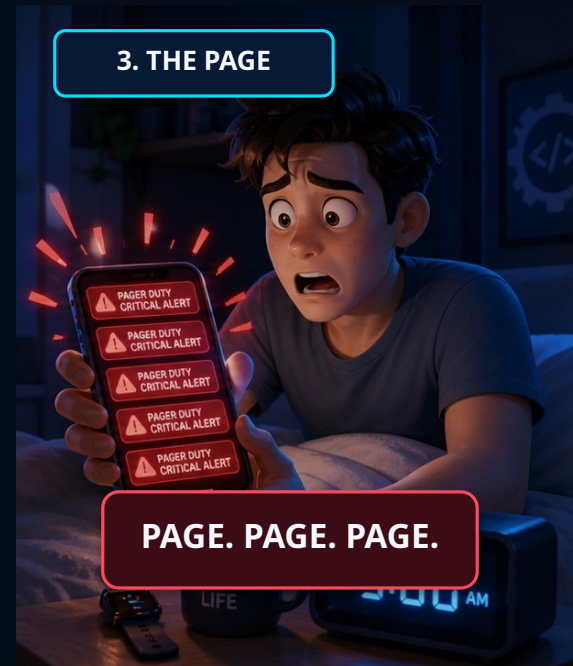
1. THE BUG



2. THE FIRE

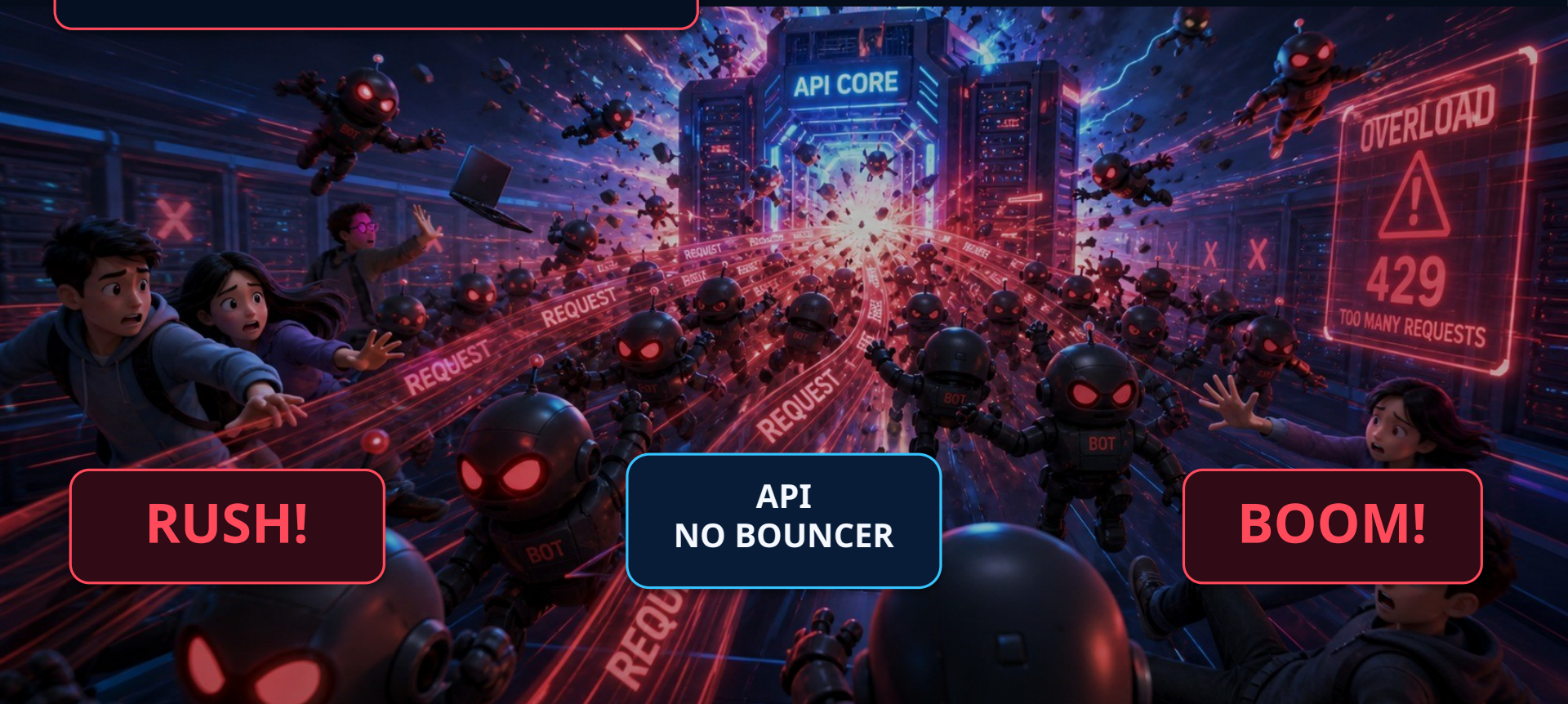


3. THE PAGE



No attacker. Just a typo.

WITHOUT LIMITER...



RUSH!

**API
NO BOUNCER**

BOOM!

Every API without limits eventually meets the same fate.



WHAT GETS HAMMERED:

Login endpoints — credential stuffing
Signup — fake account farms
Search — scrapers & data theft
Public APIs — quota abuse

Only ¼ of the story.

2

ACCIDENTAL DDoS



MOST OUTAGES AREN'T ATTACKS:

Mobile app retrying forever
Cron job stampede at midnight
Viral tweet → 1M users in 60 seconds
Cache miss → DB thundering herd

Your own users break you.



ONE LOUD USER STARVES THE REST:

One user = 80% of all DB CPU
Free-tier swamps paid customers
Slow query blocks the whole pool
SLA dies for everyone else

Your biggest fan = your worst enemy.



EVERY REQUEST COSTS MONEY:

GPU calls — \$\$ per LLM token
DB queries — connections, IOPS
Egress bandwidth — per GB
3rd-party APIs — per call

Limiter = your budget guard.

REMEMBER THESE FOUR.

MEMORY

Interviewers ask WHY before HOW.

1 ABUSE



bots, scrapers, brute force

2 ACCIDENTS



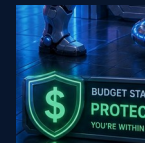
your own buggy clients

3 FAIRNESS



one whale starves all

4 COST



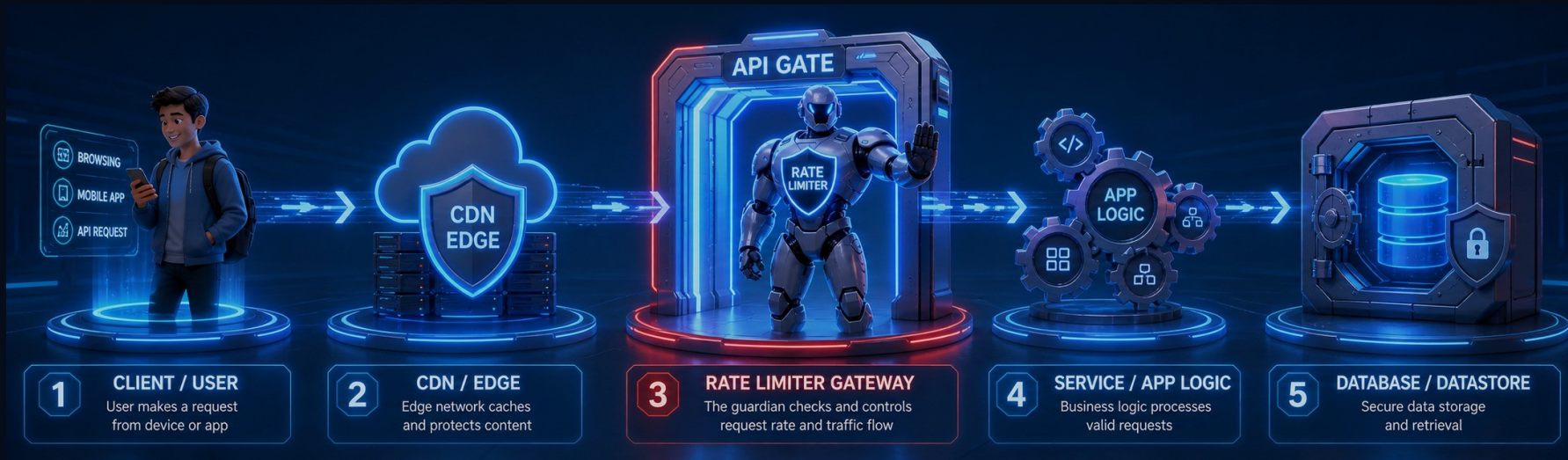
every call costs money

ABUSE · ACCIDENTS · FAIRNESS · COST

WHERE DOES LIMITER STAND?

ARCHITECTURE

4 places he can guard the door



CLIENT

the user

CDN

edge

GATEWAY

the usual spot

SERVICE

your app

DB

data store

← MOST TEAMS PUT HIM HERE

You can put him at any layer — and that choice has trade-offs.

EARLIER = CHEAPER. LATER = SMARTER.

TRADE-OFF

CDN

GATEWAY

SERVICE

DB

cheap / blunt

smart / expensive

CDN

✓ Cheapest. Stops attacks at the edge.

✗ Blunt — cannot tell users apart.

GATEWAY

✓ Knows user, JWT, tenant.

✗ Adds 1–5ms per request.

SERVICE

✓ Smartest — knows business context.

✗ Already paid for the request.

DB

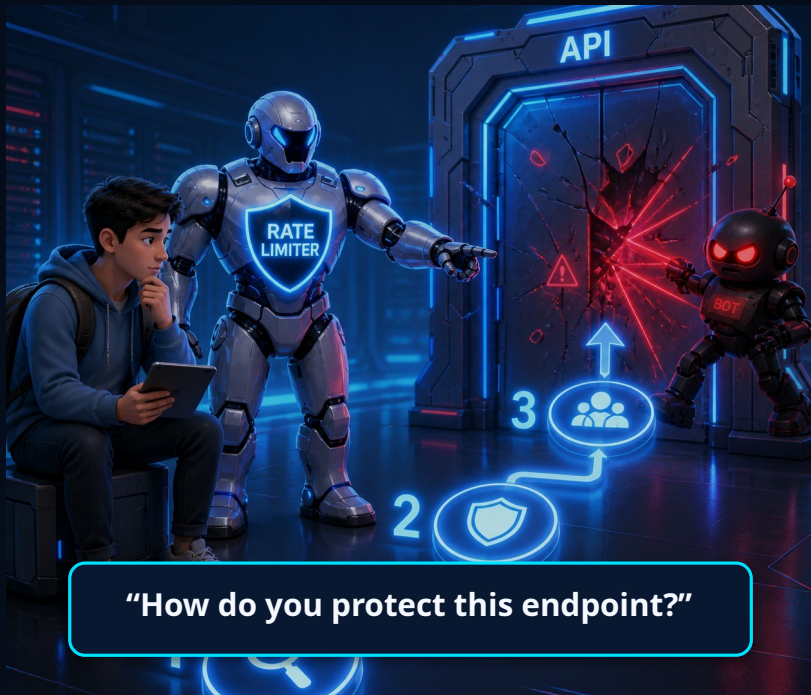
✓ Last line of defense.

✗ Most expensive — DB already engaged.

This trade-off line is what interviewers want to hear.

THE INTERVIEW HOOK

INTERVIEW



DON'T JUMP TO ALGORITHMS.

Walk this path:

1

WHY

abuse · accident · fairness · cost

2

WHERE

CDN · gateway · service · DB

3

HOW

algorithm — next video

Most candidates start at step 3. Don't.

RATE LIMITER FUNDAMENTALS

SAVE THIS

CHEAT CARD · EP 1

WHY?

Abuse · Accidents · Fairness · Cost

WHERE?

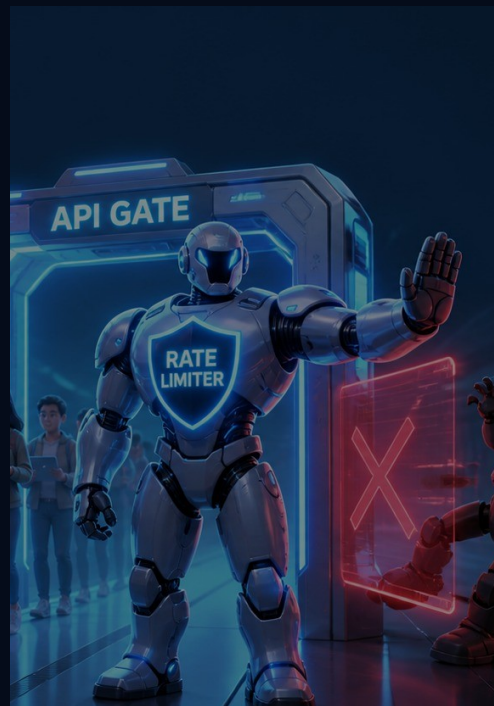
Client → CDN → [GATEWAY] → Service → DB

SAY:

“Earlier is cheaper. Later is smarter.”

PATH:

WHY → WHERE → HOW (never start at HOW)



Screenshot → 4 more cheat cards across the series.

NEXT:

THE 5 ALGORITHMS, AS COMICS.

Token Bucket · Leaky Bucket · Sliding Window...

